

Jagraj Gill

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Education

Simon Fraser University – Bachelor of Science in Computing Science

Dec 2027

Skills

Languages: C, C++, Python, Java, JavaScript, TypeScript, HTML, CSS, x86 Assembly

Frameworks: React, Node.js, Express.js, REST APIs, GraphQL, FastAPI, CI/CD, Agile

Databases & Tools: Git, Docker, MongoDB, MySQL, SQLite, AWS S3, Linux, ESP-IDF

Embedded Systems: STM32, ESP32, ARM Cortex, RTOS, I2C, UART, TCP/IP

Experience

Software Engineer Intern, Savi Finance – Toronto, ON

May 2025 – Dec 2025

- Developed an AI-powered task scheduling system with React and MongoDB that ranked tasks by priority, reducing planning time by 10%
- Built shared modules across React and React Native with a MongoDB backend for a fintech SaaS platform serving 1000+ users, shortening release cycles by 15%
- Integrated an image upload pipeline using AWS S3 and GraphQL, adding custom preview handling, persistent file storage, and error-state validation to reduce image load failures by 10%
- Improved release reliability by building GitHub Actions CI/CD workflows and React test cases to catch regressions before deployment

Projects

AI-Powered Study Assistant – Python, JavaScript, FastAPI, Gemini API

github.com/jagrajg/AI-Study-pal

- Built an LLM-powered pipeline that converts raw PDF documents into structured flashcard datasets, automating study material generation from unstructured text
- Improved AI output consistency by designing a strict JSON schema for reliable flashcard parsing
- Implemented validation logic in Python to verify data structure, field integrity, and content quality, preventing malformed or incomplete AI responses from entering the system
- Improved study workflow by developing an interactive JavaScript flashcard interface with flip animations and navigation controls, allowing users to review AI-generated cards in real time

Smart Environmental Monitoring System – C, JavaScript, SQLite, FastAPI

github.com/jagrajg/ESP32-IoT

- Delivered real-time sensor visibility by integrating ESP32 firmware, FastAPI REST endpoints, SQLite storage, and JavaScript visualizations into a full-stack monitoring system
- Developed a FastAPI backend that ingested structured sensor data from ESP32 firmware over Wi-Fi, storing readings in SQLite and serving live updates to a JavaScript dashboard
- Enabled 24-hour trend tracking by building a SQLite data pipeline with REST endpoints for live sensor readings

Portfolio Performance Tracker – Java, Swing

github.com/jagrajg/InvestmentTracker

- Built a Java Swing application to visualize portfolio growth across multiple assets and holdings
- Calculated portfolio performance by processing asset prices, quantities, and purchase history across holdings
- Improved input reliability by adding exception handling and validation for dynamic portfolio entries

Wireless Robotic Arm – C, ESP32, Embedded Systems

github.com/jagrajg/ESP-32-Arm

- Built the full software stack for a 4-DOF robotic arm on ESP32, spanning embedded C firmware, analog signal processing, and wireless communication between controllers
- Developed embedded C firmware that processed analog sensor inputs and drove PWM signals across 4 servo joints for real-time actuator control, enabling synchronized multi-joint robotic movement
- Implemented ESP32 wireless communication to synchronize robotic motion through live data transmission
- Structured modular control logic to separate input processing, wireless communication, and actuator control